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Ensemble Machine Learning based Weather Prediction System

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Abstract—The technique of predicting the weather using science and technology for a particular area is known as weather forecasting, we employed the ensemble approach to produce more precise results. Ensemble learning enhances machine learning outcomes by mixing numerous models. The predictive performance of this strategy is superior to that of a single model. Many machine learning methods and algorithms can anticipate changes in the weather. According to prior studies, there are many additional weather forecasting methods. Depending on various factors like temperature, wind, humidity, and rainfall.

Keywords— Ensemble learning, SVM, k-NN, XG Boost, Weather, Machine Learning, Rainfall, Forecasting Methods

I. Introduction

Various sensors in meteorological data for weather forecasting collect a massive amount of data. Gathering such large amounts of data is a critical process that helps people's daily lives. To prepare for potentially catastrophic weather conditions, humans need to be able to predict the weather. Other fields, including business, agriculture, tourism, and decision-making, can also benefit from it. Ensemble learning, as previously said, enhances machine learning outcomes by integrating many models[1]. When compared to a single model, this approach produces better predictive performance. Ensemble learning is classified into two types bagging and boosting. These two combine estimates from several models to lower the variance of a single estimate. A more reliable model might as a result, develop. Classification of weather forecasts also makes use of machine learning techniques. There are ML approaches that forecast weather factors like temperature, humidity, wind, and precipitation. A machine learning algorithm known as an ensemble-based technique is constructed by combining various, the Nearest Neighbour (KNN), SVM, the Gradient Boost technique, Decision tree, Boosting (XGBC), and Extreme Gradient are examples of machine learning algorithms and other. The ensemble-based approach can improve prediction accuracy and yield better outcomes[2].

The capacity to effectively anticipate the weather is particularly significant in many different sectors of modern life, including but not limited to agriculture, transportation, emergency planning, and urban growth[3]. Accurate and timely forecasts are very important in order to fulfil the goals of making well-informed decisions and mitigating the potentially detrimental impacts of weather-related events. Ensemble learning has been a useful approach for improving weather forecasts in recent years. AI significantly causes this trend[4]. This tendency is due to rapid AI breakthroughs. This study examines ensemble learning-based weather prediction system deployment and architecture. Using many models improves weather forecasting.Classical weather prediction methods use complex math and empirical data to forecast weather. Get the most detail. Although these models are more accurate, they still have limitations and uncertainties[5]. The groundbreaking ensemble learning method combines model projections to create more accurate and comprehensive predictions. This ensemble approach employs models to represent several components of the complex atmospheric system.

Collective knowledge of numerous models may surpass any single model in ensemble learning. Combining model findings into a single prediction increases accuracy, lowers biases, and enhances forecast reliability in the ensemble system[6]. Classical models worry about overfitting, but ensemble learning tackles it. This is done using several model outputs.Ensemble learning may be used in numerous ways for weather prediction. This entails selecting meteorological variables, data assimilation techniques, and numerical models. After statistical correction and uncertainty quantification, ensemble methods may increase prediction accuracy. Ensemble methods enable this.In addition, machine learning in the ensemble structure lets the system adapt to external changes. This is feasible because the ensemble structure uses machine learning. Machine learning algorithms can update real-time forecasts. Technology can detect short-term weather changes, boosting forecasts[7].To assess its effectiveness and

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utility, we test an Ensemble Learning-Based Weather Prediction System against historical weather data. This study examines ensemble size, variety, and training data on system predictions. The system's ability to forecast and control severe weather will also be studied.Weather prediction technologies using ensemble learning have improved meteorology. Combining models may improve weather forecasts' accuracy, reliability, and resilience. We want to learn more about the atmosphere and predict its behaviour from this investigation. This study may assist numerous organisations and businesses that utilise weather data for planning and decision-making.

II. Literature Survey

Our model incorporates the findings of the Random Forest (RF), Gradient Boosting Decision Tree (GBDT), Naive Bayes Bernoulli (NBB), and K-Nearest Neighbours (KNN) algorithms. To enhance the precision, the ultimate forecast is combined using maximum voting [1]. The results of previous algorithms are being considered.

(a) Random Forest: The Random Forest decision tree ensemble approach classifies meteorological data using many subtrees. Each tree's classification findings are pooled after manufacturing. Random forests resemble decision trees[8].

(b) GBDT: This algorithm incessantly tampers with the weight sample of the module while concurrently assembling a sizeable number of erroneous models. The GBDT ensemble boosting strategy entails continually creating decision trees that are dependent on the value that has been updated based on the regression[9]. A newly built regression decision tree is utilised at each step of the project in order to compensate for any mistakes that may have happened in the original classification of the input meteorological information. This is done in order to make sure that the project is completed successfully. The use of GBDT is advantageous due to the fact that it controls non-linear complex variables, which is an advantage in and of itself. It does not suffer from the effects of overfitting, and as a result, it performs far better when being trained than it does while being assessed.

(c) The use of the naive bayes method will lead to an increase in the precision of weather forecasting analysis as a consequence of the improvement[10]. The design and execution of this strategy are both grounded on the idea of occurrence probability as its primary foundation. In addition to this, it provides an accurate result despite using a simple approach and a characteristic of the weather dataset. This is particularly impressive given that it just makes use of these two factors. The NBB model makes use of the Bayes theorem, and one implementation of the theorem may be explained as follows:

Probability(X|Y) equals P(Y|X) multiplied by P(X) divided by P(Y) (d). KNN: Both difficulties with making accurate predictions and those with classifying the upcoming weather may be tackled with the help of the KNN approach. It does this by doing an analysis on the weather vector values that have been provided for the purposes of forecasting and observation[11]. This allows it to produce new data points. It takes a few pieces of input data and compares them to many values that are the closest neighbours to arrive with an accurate prediction of the current state of the weather. The distance function for the missing weather characteristic is used to assess the values, and the evaluation is based on how similar the characteristics are to one another. The distance function for the missing weather characteristic is used to evaluate the values.

[2] Ensemble techniques make use of a number of distinct learning algorithms to deliver a higher degree of predicted accuracy than any one of the individual learning algorithms could probably provide on its own. This is accomplished by combining the results of many learning algorithms into a single model[12]. The results of numerous different learning algorithms are combined in order to achieve this. A small selection of the ensemble and prediction algorithms that have been used includes the Support Vector Machine, the Random Forest, the Linear Discriminate Analysis, and the Generalised Boosted Algorithm.

(a) Linear discriminant analysis, which is sometimes referred to simply as LDA in certain circles, is a method that may be used to reduce the overall number of dimensions. Pattern alignment is a term that refers to one of the pre-processing steps involved in machine learning. Its goal is to bring the advantages of high-dimensional space down to low-dimensional space, so avoiding the issues that are connected with dimensionality while simultaneously reducing the amount of resources that are required and the expenses associated with dimensions[13]. In order to reduce the total number of measurements, LDA concentrates its major attention on accurately forecasting the highlights over larger measurement zones. There are three steps that need to be completed before it can be finished.

(b) The Generalised Boosting Algorithm: [Note: [Note: In order to increase the accuracy of templates, generalised boosting models are often fitted to various types of decision trees. An arbitrary subset of the massive amount of data is chosen by increasing the likelihood that in the most recent prototype, data that is worthless as suggested by earlier trees will be selected[14]. This is accomplished by increasing the likelihood that an arbitrary subset of the data will be chosen with each subsequent iteration of the model, which involves a tree that increases the likelihood that an arbitrary subset of the data will be chosen. This would mean that the model would take into account any misconceptions about the tree's assumptions in order to correctly pick the next tree, and so on and so forth if the initial tree is adequate. The framework is continuously attempting to enhance its realism by taking into consideration the danger presented by trees that have been placed in the environment in the past.

(c) Support Vector is a method of artificial intelligence that is subject to regulations and may be used to situations that include classification or regression. It is designed to assist in the resolution of issues of this kind. In the SVM technique, each data point is presented in the form of a data point in an n-dimensional space, with each component estimate representing the expectations of a certain organisation[15]. Because the hyperplane was positioned in such a way at that particular

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instant in time, the two classes were effectively divided into distinct groups as a consequence. The use of decision boundaries, which are more often referred to as hyper-planes, is beneficial to the visualisation of data centres. Depending on the specifics of the situation, the data focuses that fall to the left or right side of the aircraft can be assigned to a different category. The quality of the feature that is being used also plays a role in determining the amount of the hypersurface that is produced[3].

We started our analysis of the climatological data for Chittagong on January 1, 2000 and continued it all the way through December 31, 2020, giving us a total time frame of twenty years to work with. Following that, educated guesses were made about the missing data by making use of a basic computer and the process of averaging[16]. Following that, the Lime-scale estimator was used to the dataset in order to do the scaling. once finishing the preprocessing stage, we used seven different algorithms on our dataset once it was finished being processed. There have been many different approaches used, and one of them is the usage of artificial neural networks. The following techniques are considered to be boosting approaches: gradient, Ada (adaptive), XG, stacking random forests, stacking neural networks, and stacking KNN. Stacking KNN and stacking neural networks are two further methods. Boosting is a technique to ensemble modelling in which a series of weak classifiers are combined into a single robust classifier. This is the goal of the "boosting" approach. Continue repeating this either until the maximum number of models that may be included is reached or until it is possible to make a forecast that is acceptable for the whole training data set. Neural networks, which are a branch of machine learning, serve as the basis for the deep learning approaches. Common names for these types of networks include artificial neural networks, simulated neural networks, and simulated neural networks abbreviated as SNNs. These artificial neurons, which were created to imitate the way that genuine neurons communicate with one another, took their form and their names from the human brain, which served as the foundation for both of these aspects of the artificial neurons. The third technique for machine learning ensembles is called stacking, and its objective is to find the best method for integrating the predictions given by numerous powerful machine learning models. This strategy's name comes from the fact that its goal is to uncover the best way. We observed that stacking neural networks generated a result that was much more accurate for our goal value when compared to the other strategies that were investigated.

III. Proposed System

A. Ensemble Methodology

The results of machine learning might potentially be enhanced by the use of ensemble learning, which involves combining multiple distinct models. The capability of using this strategy to forecast outcomes in a reliable manner is much enhanced when compared to the use of a single model as the predictive tool. A group of people who will be voting as classifiers must first go through training. Only after that can they really cast their ballots. The major advantage of using an ensemble is that it enhances the accuracy of the prediction. Boosting, stacking, and bagging are the three primary subcategories that fall under the umbrella of ensemble learning approaches.

Boosting- Aims to create a powerful classifier by combining weak learners. Weak models are applied in series to build a model. The training data set is first used to create a model. The second is then created to solve the first's flaws. Model addition keeps going until either the maximum number of models can be added or all of the training data can be correctly predicted.

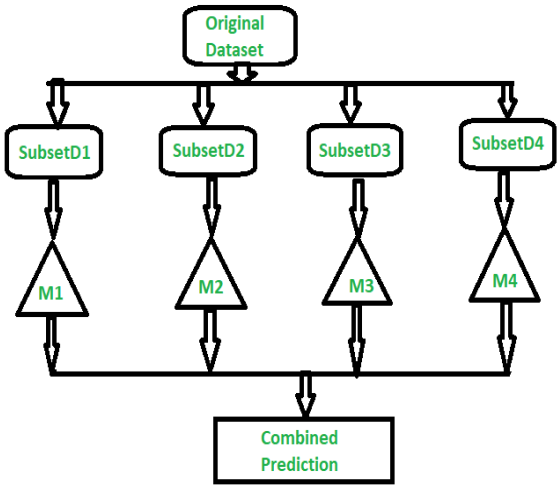

Fig. 1. Ensemble Flowchart

- Bagging- Bagging is a technique that is used to improve the accuracy and consistency of machine learning algorithms that are used in statistical regression and classification. This method is quite good in reducing variation and also assists in preventing overfitting. In approaches using decision trees. Bagging is an important part of the model that is known as the model average, and it is widely employed.
- Stacking - A well-known ensemble modelling strategy in machine learning is stacking, where several weak learners are grouped together in parallel so that we may make stronger predictions for the future by combining them with Meta learners. A stacking algorithm uses the sub=model outputs as input and tires to determine how to best combine the input prediction to achieve a better output prediction. Stacking, also known as stacked generalization, is a more advanced version of the model-Based Average ensemble method with equal contributions from each sub model based on their performance weights to produce a new model with improved predictions

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B. *Proposed Methodology*

The whole of the recommended framework has been established, including the gathering of meteorological data, its preliminary processing, feature selection, the creation of ensemble-based evolutionary models, and the evaluation of the outcomes of the forecasts. The technique of predicting that has been proposed is broken down into its component parts and shown in a figure with the label "Fig. 2."An sophisticated kind of machine learning known as ensemble learning combines the findings of a number of different machine learning algorithms into a single set of findings. When compared to the usage of only one machine learning algorithm, this will result in improved weather forecasting predictions.

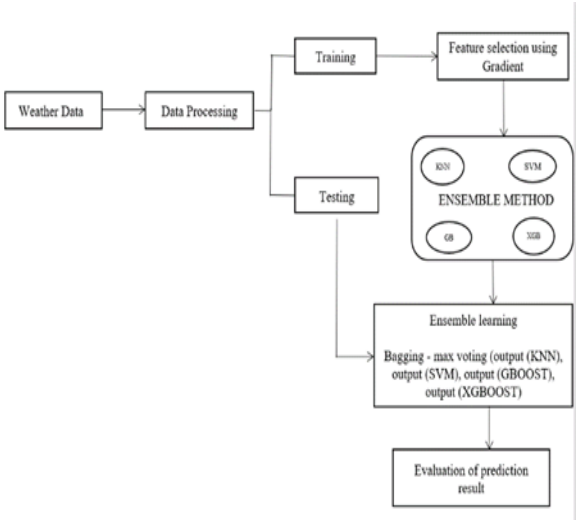

Fig. 2. Weather Forecasting Framework

Training and testing are included in the weather data. The data are preprocessed in order to normalise them and fill in any gaps. The gradient is used to identify characteristics. The chosen attributes are categorised via the use of the Extreme Gradient Boosting (XGBC), Support Vector Machine (SVM), K-Nearest Neighbour (KNN), and Gradient Boosting Method. This allows for accurate forecasting of the weather. The aforementioned algorithms are combined using bagging, also known as max-voting, in order to reach the final results that were predicted.

IV. Result and Execution

The goal of ensemble techniques is to improve the accuracy of models by combining many different models rather than depending just on a single model. The accuracy of the findings may be greatly improved using the integrated method because to the combination of various models. The K-nearest approach achieves an average accuracy of 75%, Extreme Gradient Boosting achieves an accuracy average of 83%, Support Vector Machine achieves an accuracy average of 77%, and Gradient Boosting achieves an accuracy average of 62%.As a consequence of this, the ensemble approach achieves a combined accuracy score of 90% while maintaining a low error rate.

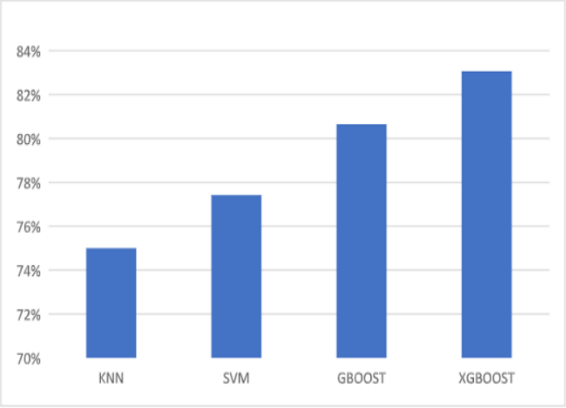

Fig. 3. Algorithm Comparison

The following stage, which comes after importing the data into a Pandas Data Frame, is data exploration. During this step, you should look for any missing values, outliers, or skewed data. Create the dataset by first generating the dependent as well as the independent variables. Reduce the complexity of the scenario by setting the levels of the independent variables where they should be. To encode any category variables, it is advised to make use of either the Label Encoder or the One Hot Encoder. Both the training set and the test set should be derived from the dataset. Independently utilising the training data, train the Gradient Boosting, XG Boost, SVM, and KNN models, in addition to the SVM model.

In order to make predictions based on the outcomes of the testing, apply each model to the data that was collected. When applied to the early predictions, the ensemble approach makes it possible to combine those estimates into the final forecasts. Utilise measures like as accuracy, F-1 score, recall, and precision in order to conduct an analysis of the performance of the ensemble model. One strategy to improve the accuracy of the predictions produced by the models is to make adjustments to the hyperparameters that they use. First, determine which model produces the best overall outcomes, and then use that model to make forecasts based on newly collected data.

V. Conclusion

In summary, this investigation utilises Ensemble learning to anticipate the weather by merging data from KNN and SVM models. Extreme Gradient Boosting, commonly referred to as Gradient Boosting and Extreme Gradient Boosting (XGBC), is a kind of gradient enhancement technique. The present machine learning algorithms for weather prediction are assessed based on significant performance criteria, and the results of these assessments are contrasted with the help of the suggested ensemble model. When it comes to trying to anticipate the weather, we found the intriguing conclusion that an ensemble-based prediction system produces the most accurate results. Ensemble techniques make use of a variety of distinct learning algorithms in order to improve the accuracy of the models that they produce via increased prediction. One strategy for increasing productivity while also enhancing the

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reliability of weather forecasts is to make use of the newly developed ensemble model. The study that is being given has many objectives, one of which is that it should have a high accuracy rate while also having a low error rate and a fast forecast time. As a direct result of this, the ensemble method obtains a cumulative accuracy score of 90% while keeping the error rate very low.

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